

Domain Classes Identification using the Noun Technique

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Step 1: Extract All Nouns

Raw Nouns
Admin, teacher, student, parent, system, account, user, login, grade, report, attendance, class, subject, announcement, page

Step 2: Remove Irrelevant Nouns

Noun	Keep?	Reason
page	No	UI element
login	No	Authentication detail
system	No	Not a domain entity
information	No	Too generic
performance	No	Too abstract

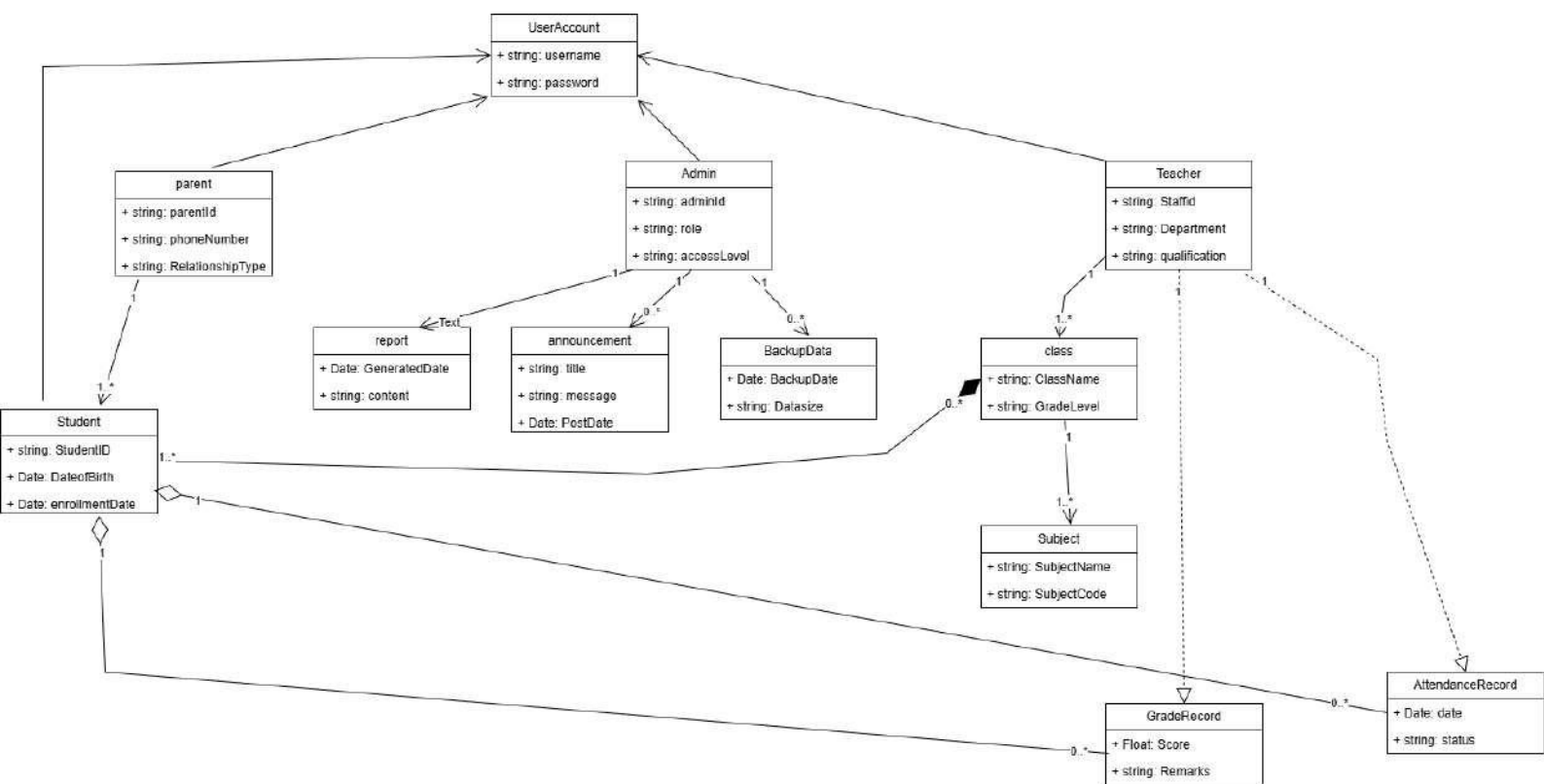
Step 3: Classify Nouns

Category	Nouns
People / Actors	Admin, Teacher, Student, Parent
Objects / Data Elements	UserAccount, Attendance, Grade, Report, Announcement, Subject, Class, BackupData
Relationships	Child (Parent → Student)

Step 4: Final Domain Classes

Final Domain Class	Description
UserAccount	Manages authentication and identity
Admin	System administrator
Teacher	Records attendance and grades
Student	Views grades and announcements

Parent	Views child attendance and grades
Class	Represents a school class
Subject	Academic subject assigned to class
AttendanceRecord	Stores attendance entries
GradeRecord	Stores student grades
Report	Generated by admin
Announcement	Posted updates
BackupData	System backups



2.1 Final Use Case List

The following Use Cases represent the high-level functional requirements of the system.

Use Case	Actor	Short Description	D2 Event Type
Manage User Accounts	Admin	Add, modify, view, or remove user accounts (Admin, Teacher, Student, Parent).	External
Record Attendance	Teacher	Mark student presence or absence for a class session.	External
Update Grades	Teacher	Enter, modify, and save grades for students.	State
View Grades	Student, Parent	Access and review individual or child's subject grades.	External
View Attendance	Parent, Student	Review attendance history and current status.	External
Manage Announcements	Admin, Teacher	Create, edit, and publish announcements/notices.	External
View Announcements	All Users	Read the latest school or class announcements.	State
Manage Study Materials	Teacher	Upload, update, and remove digital study resources.	External
Access Study Materials	Student	Download or view shared study materials.	External
Generate Reports	Admin	Produce simple summary reports for performance and attendance.	Temporal
Backup Data	System	Automatically perform a secure daily backup of all system data.	Temporal

2.2 CRUD Technique Validation

The CRUD technique is applied to the system's primary data entities to ensure all functional requirements are supported. This process validates that every required system action is correctly mapped to a Create, Read, Update, or Delete operation.

Data Entity	Actor	Action/Use Case	CRUD Operation	D1 Functional Requirement/Key Feature
User Account	Admin	Add a new user (Student/Teacher/Parent)	Create	"Add and update student and teacher details"
User Account	Admin	View/Search user details	Read	"Make it easier to manage student and teacher information"
User Account	Admin	Update user details	Update	"Add and update student and teacher details"
User Account	Admin	Remove a user account	Delete	(Implied by User Management)
Attendance Record	Teacher	Record initial attendance	Create	"Attendance record for the classes"
Attendance Record	All Users	View attendance history	Read	"Parents track their children progress," "Attendance record"
Grade Record	Teacher	Enter/Update grades	Create/ Update	"Teachers can enter or update grades"
Grade Record	Student/Parent	View grades	Read	"students and parents can see them"
Announcement	Admin/Teacher	Post a new announcement	Create	"Admins and teachers can make announcements"
Announcement	All Users	View/Read announcements	Read	"Admins and teachers can read announcements"
Announcement	Admin/Teacher	Edit or remove a published announcement	Update/ Delete	(Implied by announcement management)
Study Material	Teacher	Upload a new material file	Create	"Uploading study materials"
Study Material	Student	View/Download materials	Read	"students can download or view materials anytime"
Reports	Admin	Generate report data (view)	Read	"Generate simple reports for attendance and grades"

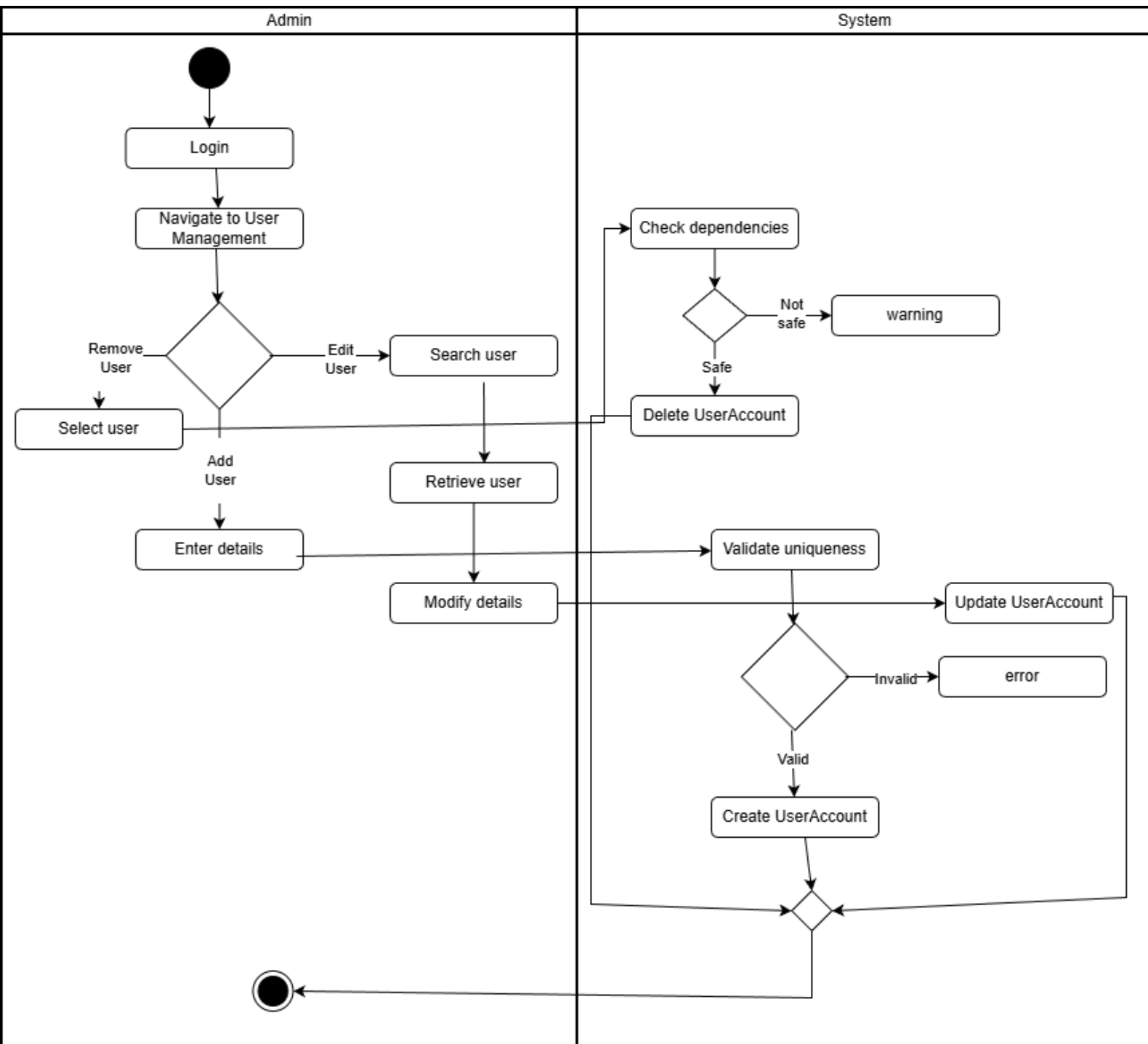
2.3 Rationale and Diagram Update

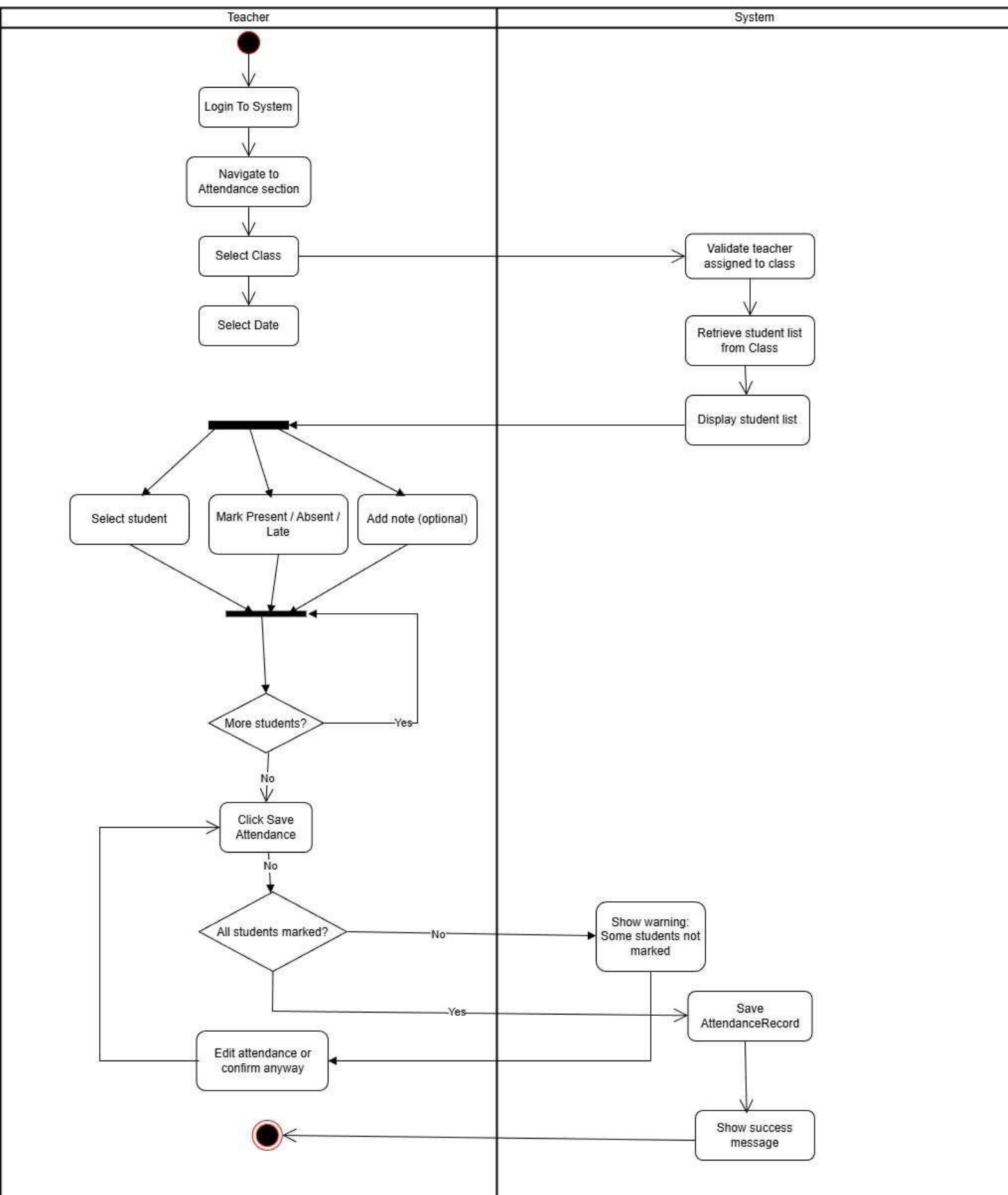
Rationale for Refinement:

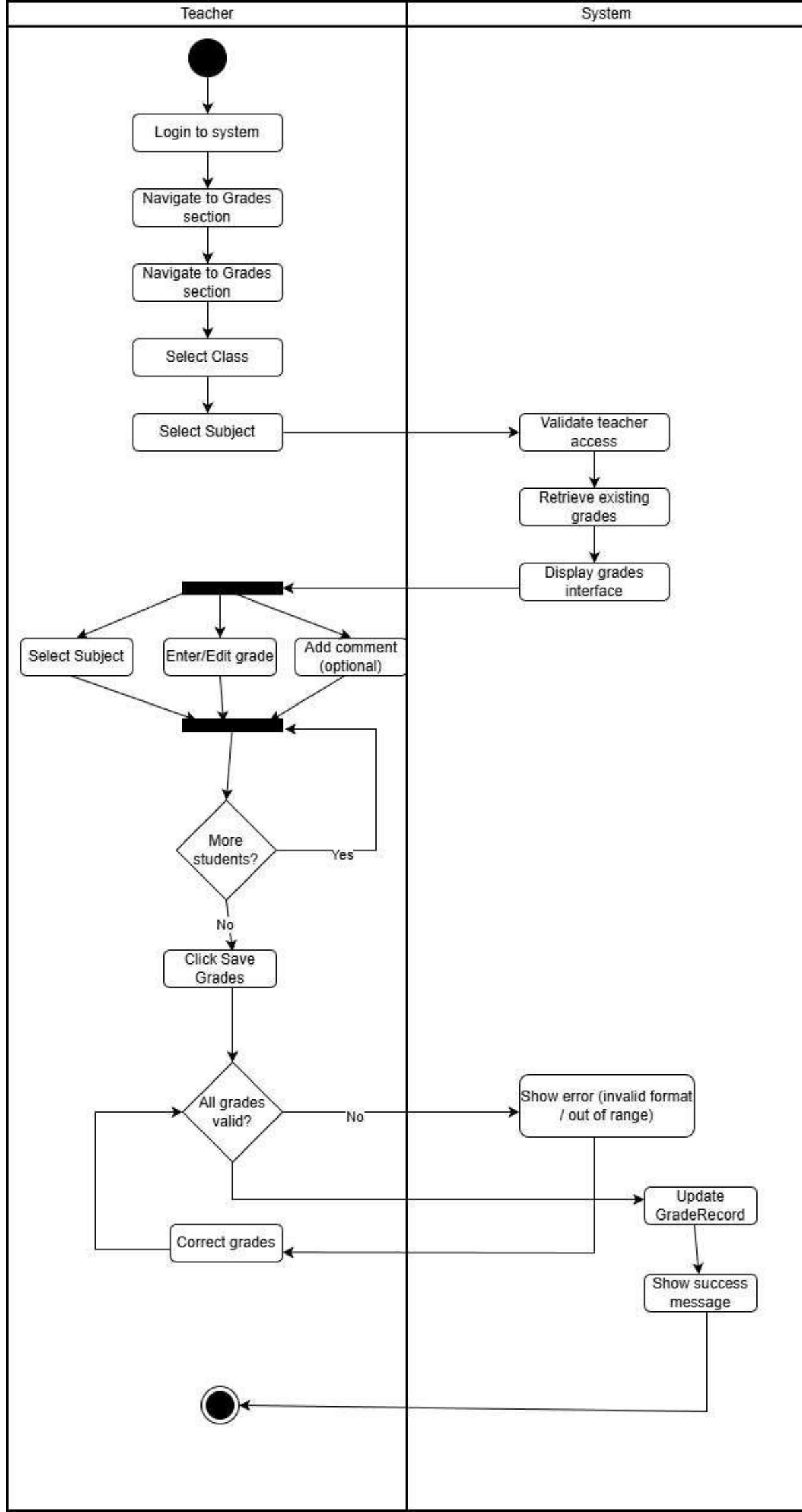
The CRUD analysis confirmed that the high-level Use Cases identified in Deliverable #2 (Manage User Accounts, Update Grades, Record Attendance) are sufficient to cover the full spectrum of required data manipulations. Two minor Use Cases were made explicit for clarity: Manage Announcements and Manage Study Materials, to fully reflect the CRUD operations related to these key content types.

UML Use Case Diagram Update:

The updated Use Case diagram below incorporates the newly refined or explicitly named use cases (Manage Announcements and Manage Study Materials) to ensure complete coverage of the system's scope.







School System

triggerScheduledBackup()

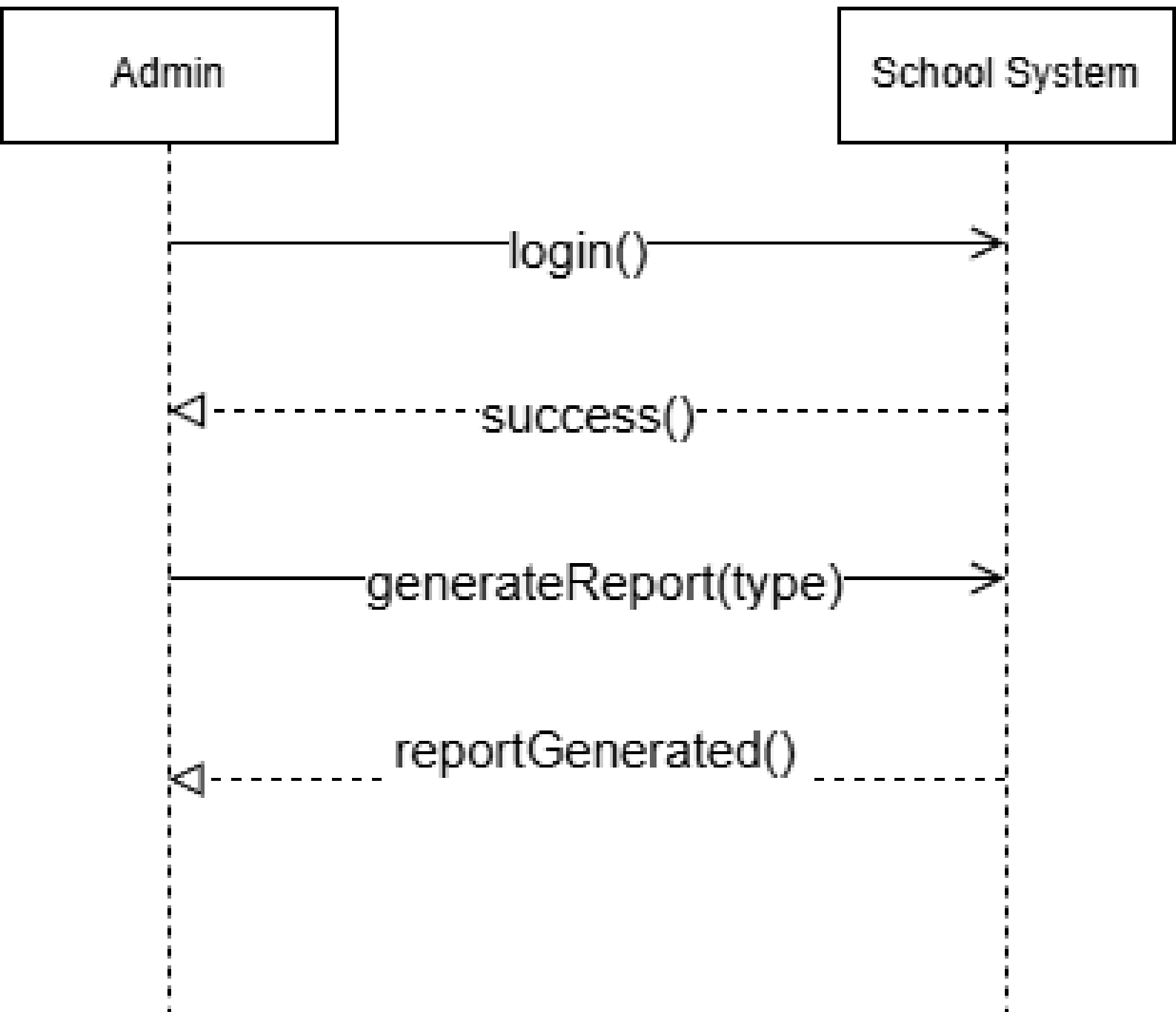


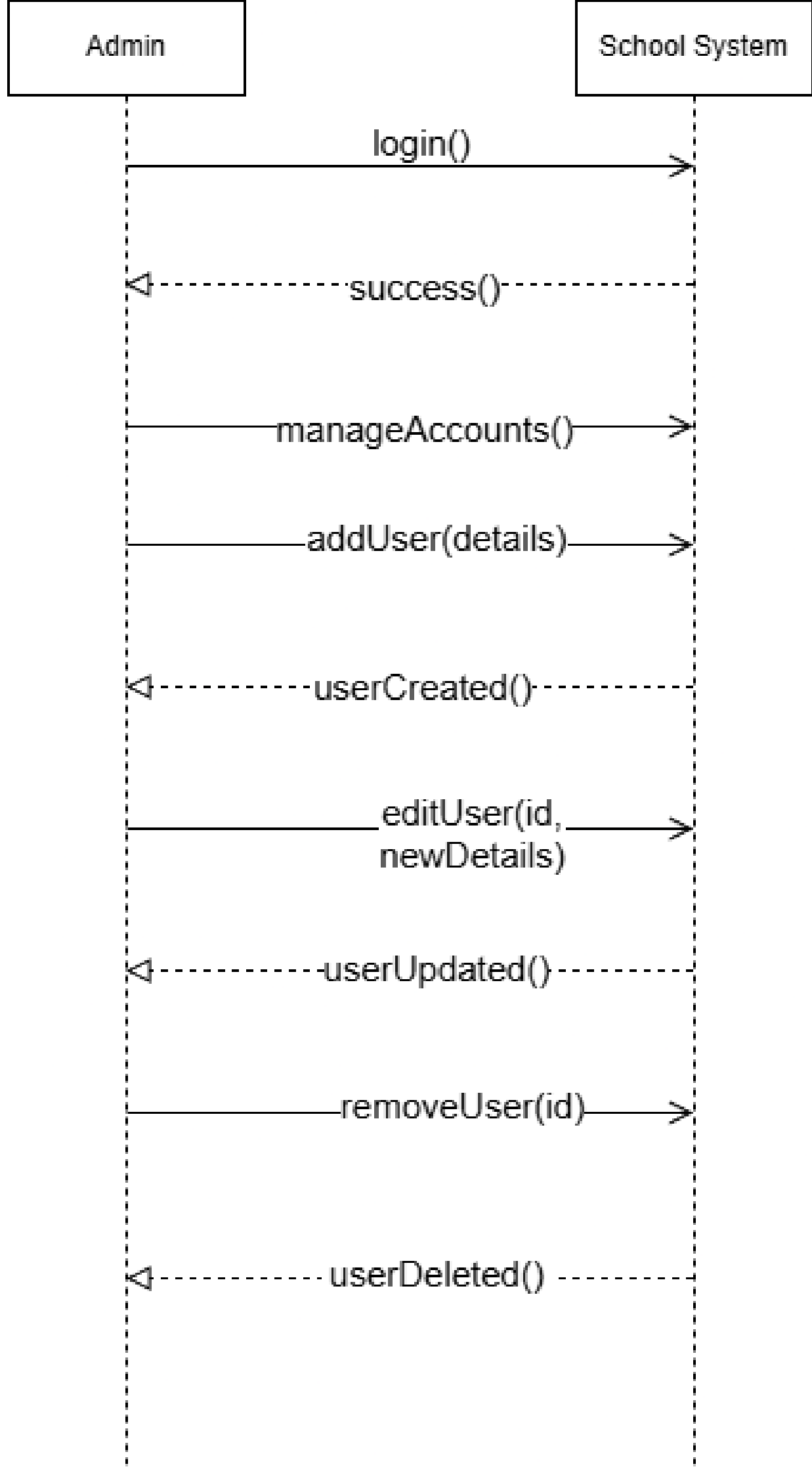
performBackup()

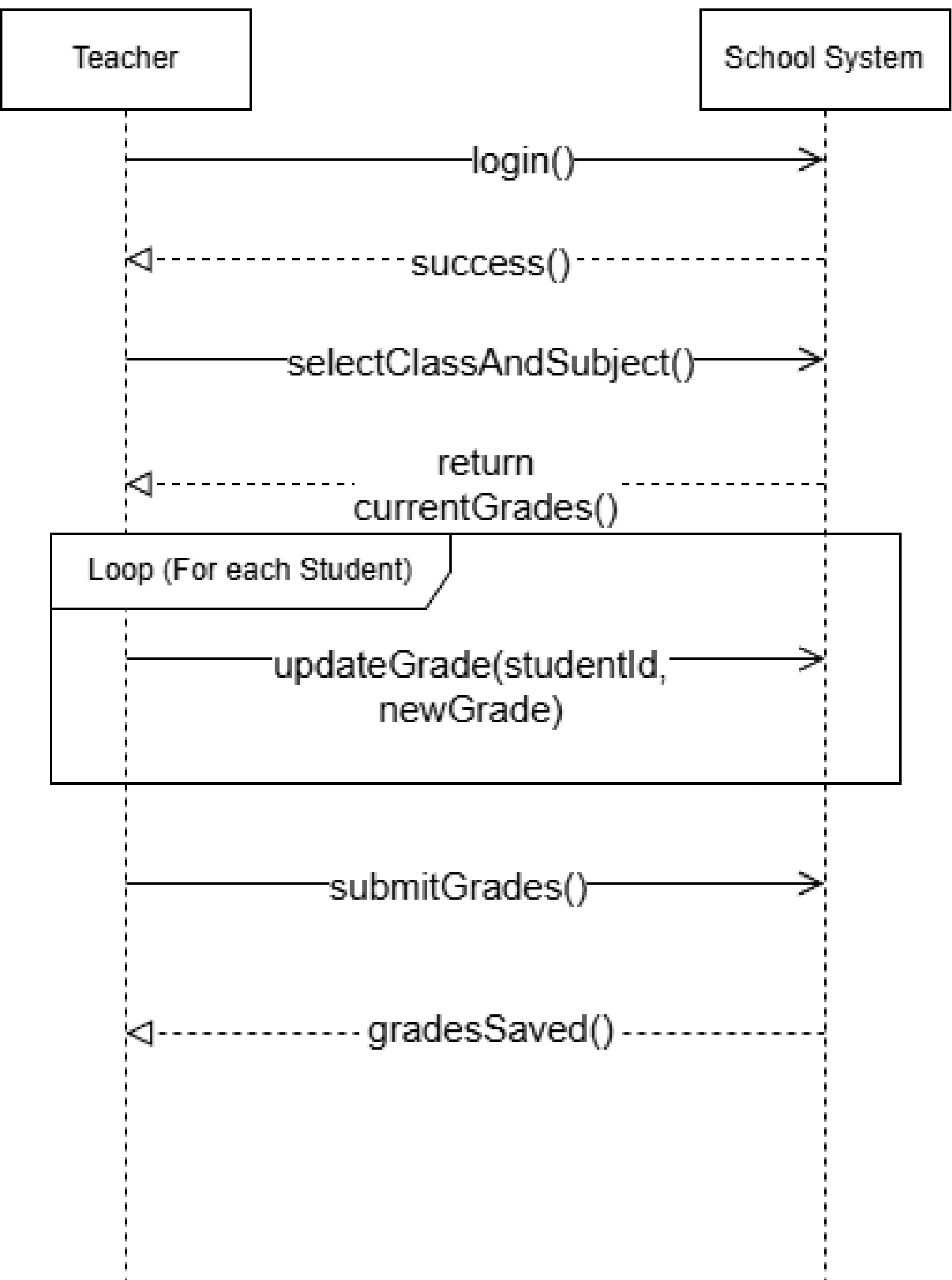


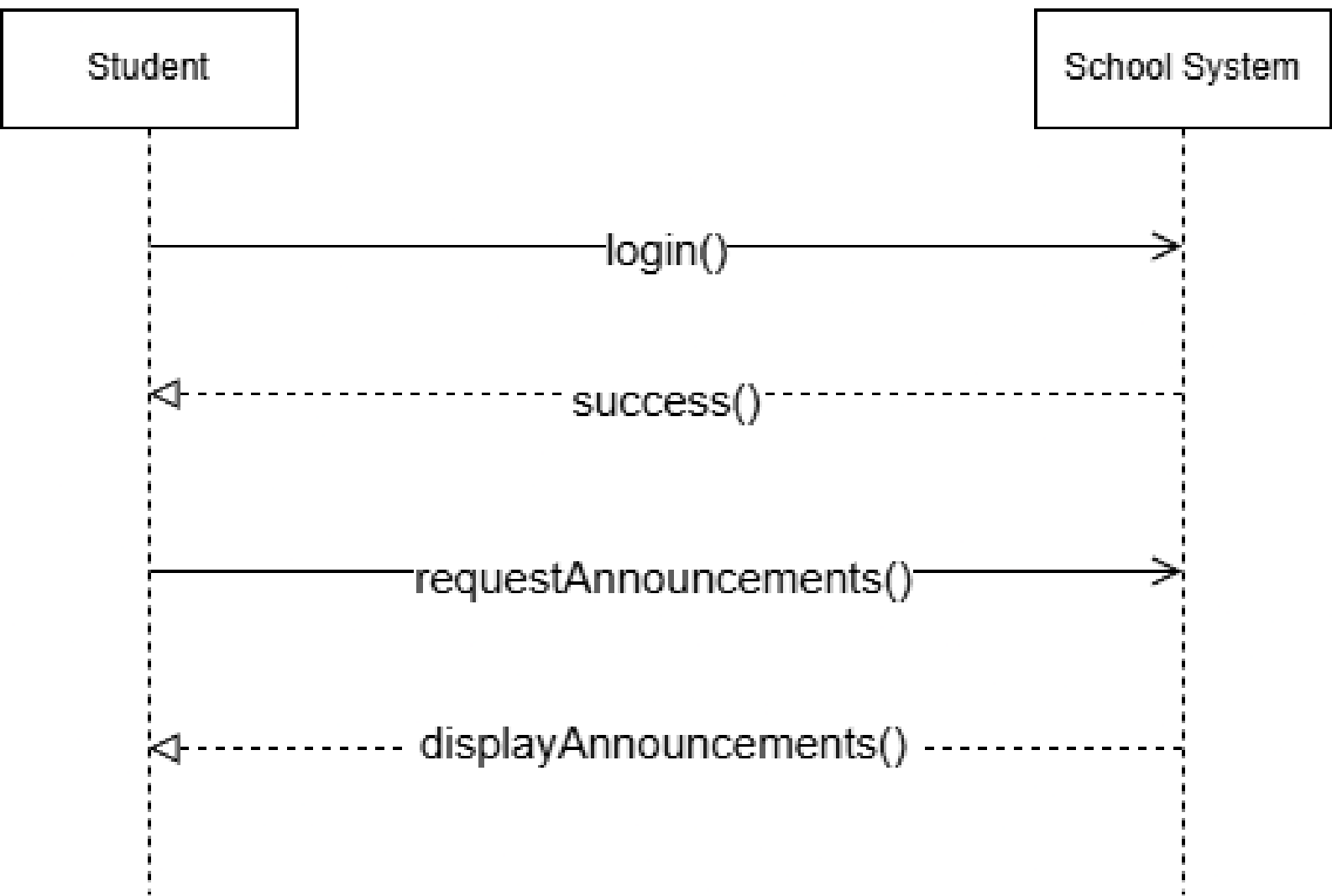
backupSuccess()

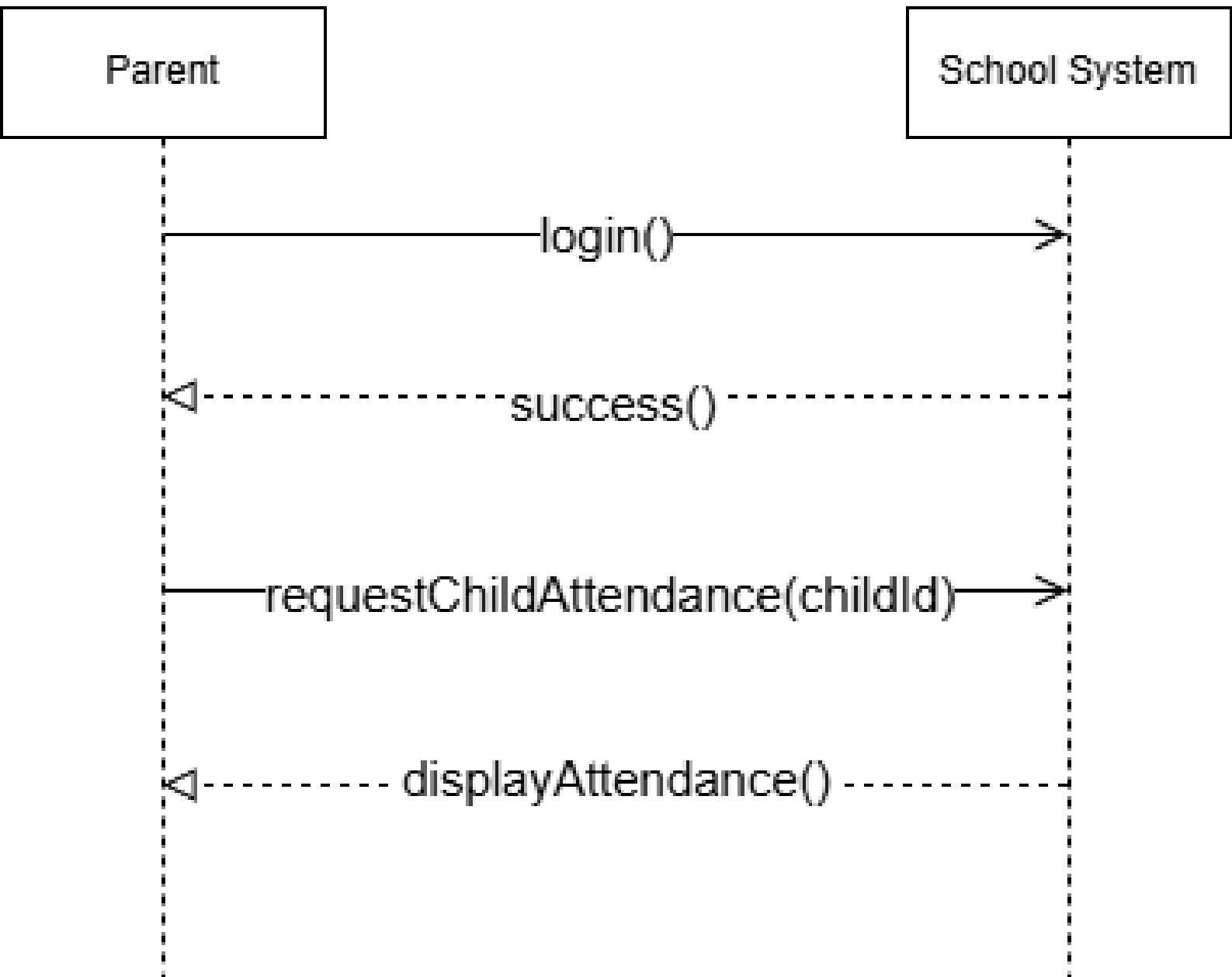


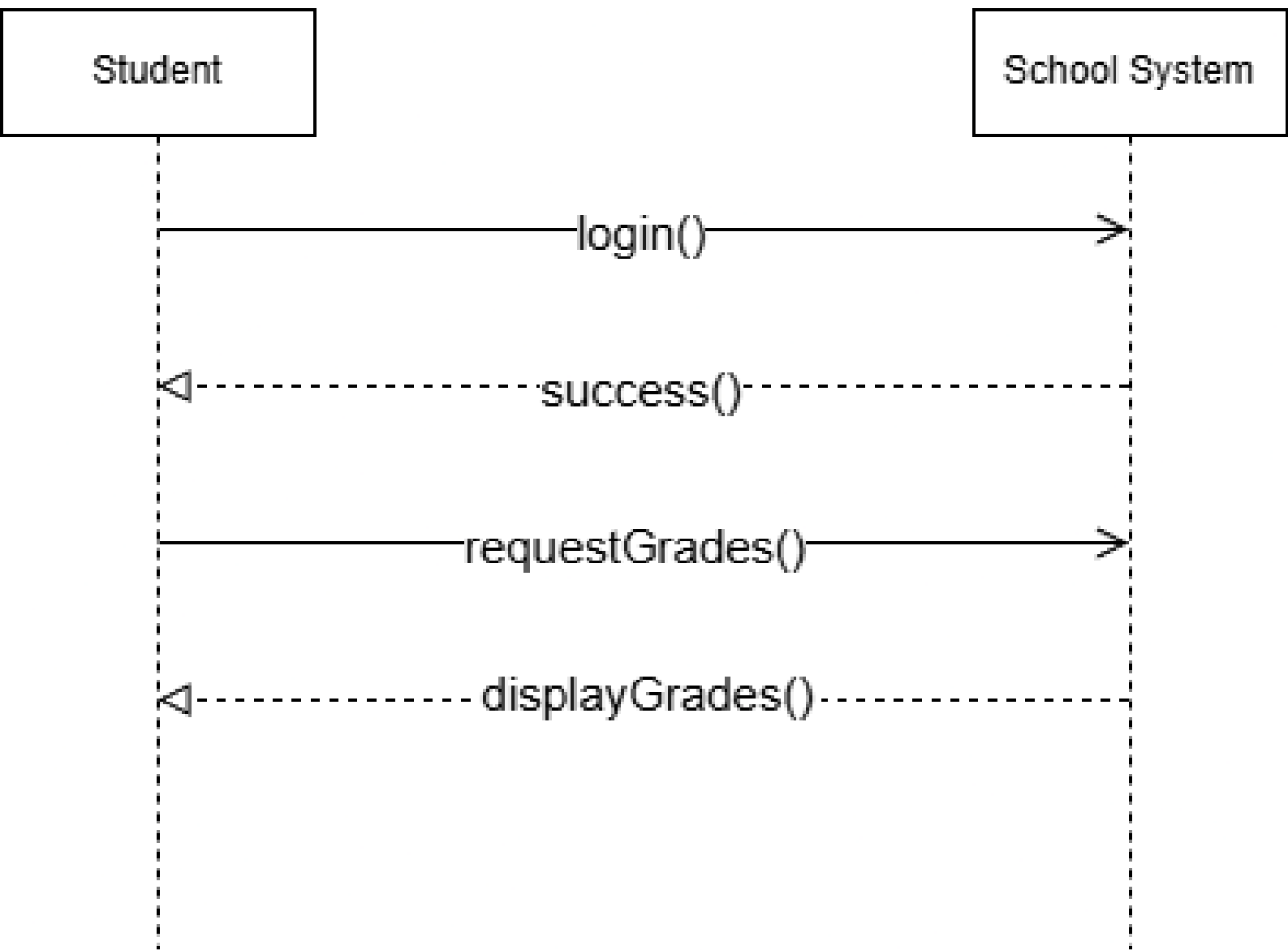


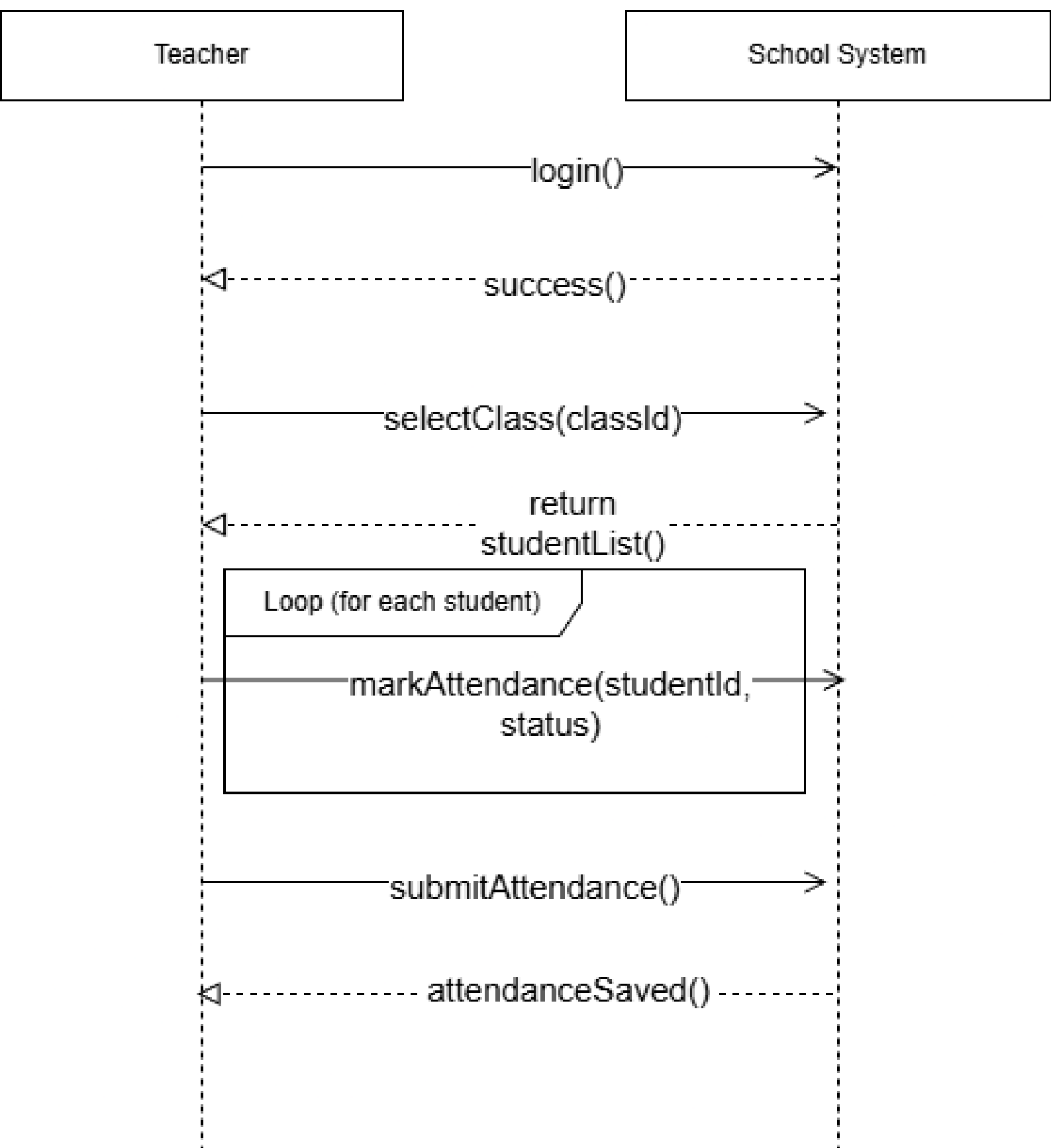


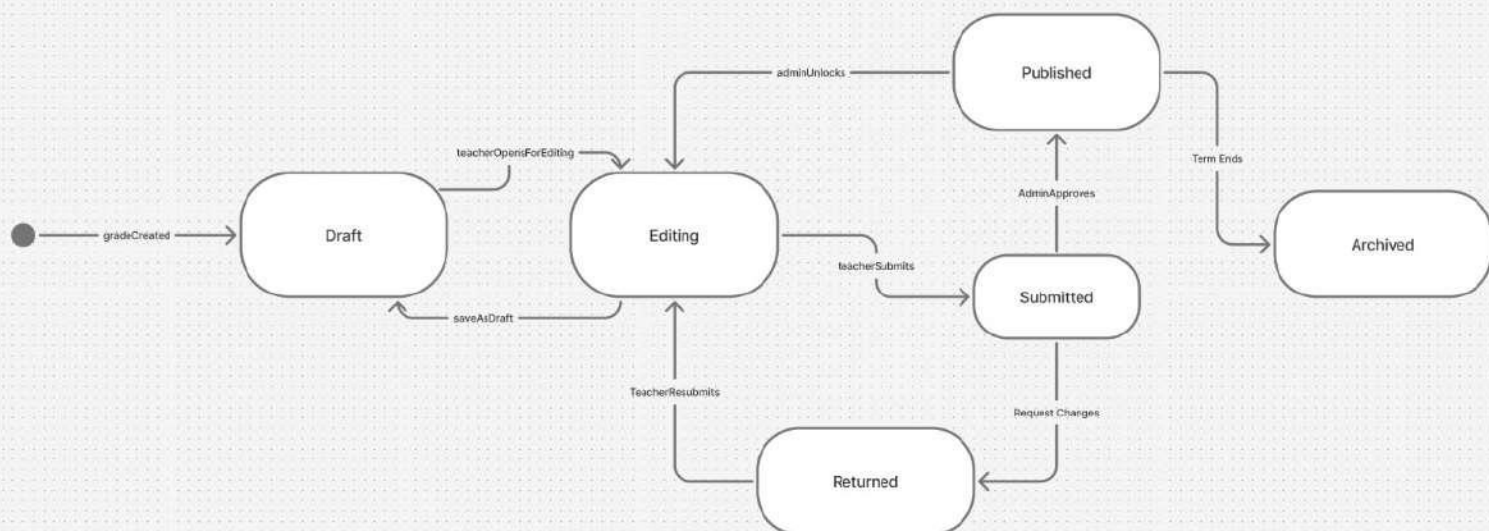


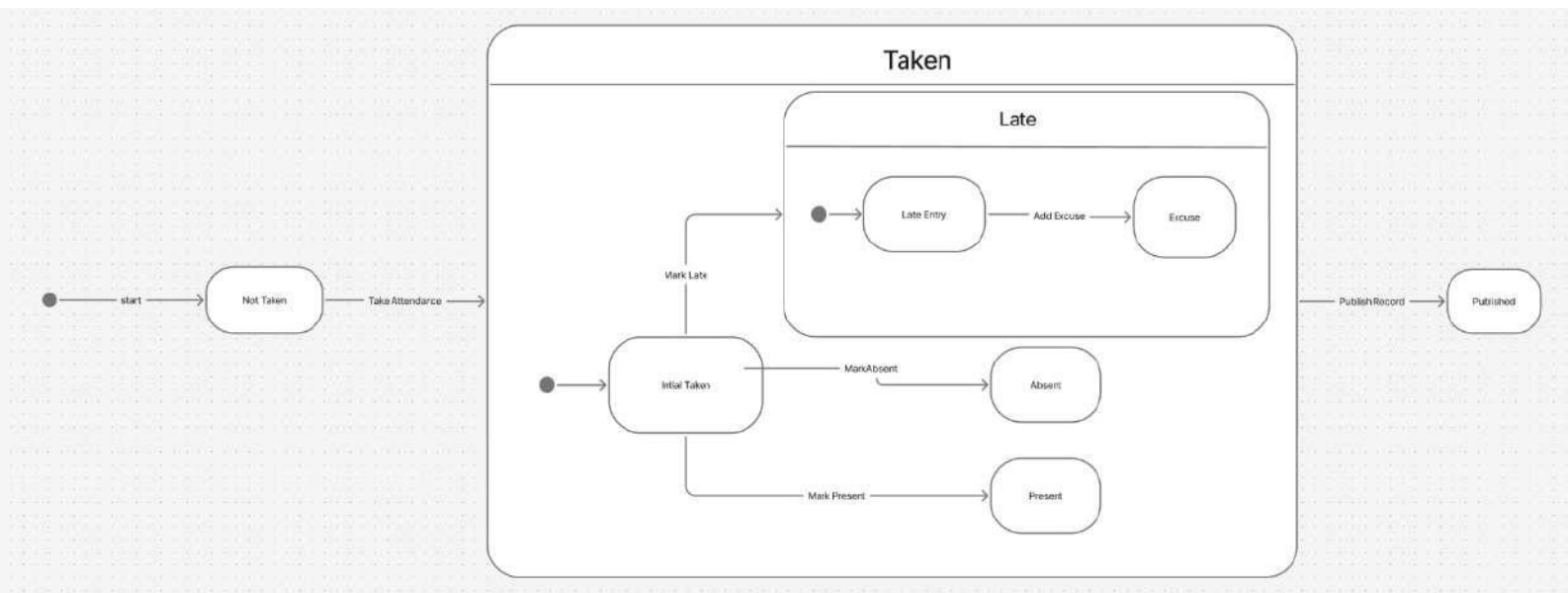












Software Requirements Specification (SRS) - School Management System

Date: November 29, 2025	Team: Marwan Khaled, Youssef Idrees, Hany Yasser, Omar Elsayed, Yassin Sameh
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The Purpose Of the Project:

The current school environment relies heavily on paper or outdated manual systems for recording student data, attendance, and grades. This process is inefficient, prone to mistakes, and results in slow communication among stakeholders. The project aims to build a centralized School Management System to digitize these core activities. The School Management System is being developed to end the inefficiency and error-prone nature of managing crucial student data through outdated paper and disjointed digital systems. Our vision is to deliver a single, unified digital hub that effortlessly connects administrators, teachers, students, and parents. By centralizing functions like attendance, grading, and communication, the system will not only save valuable time and reduce administrative burden, but it will also ensure data is organized, secure, and instantly accessible to everyone who needs it. The core value is simple: efficiency, transparency, and better communication across the entire school ecosystem.

Goals Of The Project:

The primary goal is to provide a single platform that streamlines daily school work, replacing fragmented tools .

Goal Area	Purpose/Advantage (What we gain)	Measurement (Fit Criterion)
Efficiency	To save time by reducing manual work and streamlining processes.	The system shall reduce the average time taken for a teacher to record attendance for a class from 5 minutes to under 60 seconds .
Data Integrity	To keep attendance and grades organized and easy to find in one place.	The system shall maintain an average data retrieval latency of less than 1.5 seconds for all core reports.
Communication	To help teachers and students/parents communicate better.	90% of parents and students shall confirm that they receive urgent announcements via the system within 5 minutes of posting.

The Stakeholders:

Stakeholder Category	Identified Stakeholders	Key Aspects of Interest
Executive (Client/Customer)	School Principal, Vice Principal, Head of Departments	Tracking teacher efforts, making decisions according to system data, and reviewing school progress.
Internal	School Administrator, Admins, Teachers	Data management, class schedules, attendance, grades, and uploading study materials.
Operational (Hands-on Users)	Students, Teachers, Parents	Checking grades, attendance, announcements, and accessing learning resources.
External	Parents, Software Developers	Having access to children's records; upgrading system features and security.

The Hands-On Users of the Product:

User Role	User Responsibility/Profile
Admin	Controls the whole system and manages user accounts.
Teacher	Records attendance, enters/edits grades, uploads course materials, and manages class announcements.
Student	Views personal grades, attendance history, class schedule, and school news/announcements.
Parent	Views their child's grades, attendance, and school announcements.

Project Constraints: Boundaries and Restrictions:

Mandated Constraints

Solution Constraints

- The system shall be implemented as a web application.
- The system must be accessible through a standard web browser.
- The system must support functionality across multiple devices (desktop, laptop, tablet).

Enterprise Constraints (Excluded Scope)

- The product shall not include school fees or payment systems.
- The product shall not be built as a dedicated mobile application in the current scope.

Naming Conventions and Terminology:

The following terms will be consistently used throughout the specification:

Term	Definition
User	Any individual who authenticates into the system (Admin, Teacher, Student, Parent).
Attendance	A binary record (Present/Absent) of a student's presence for a class session.
Grade	A specific mark or score assigned to a student for a subject or assignment.
Report	A synthesized view of attendance or grade data for review or tracking.

Relevant Facts and Assumptions:

Assumptions

- Internet Connectivity: All users are assumed to have an active internet connection when using the system.
 - Historical Data: Existing student and teacher data will need to be imported or migrated into the new system during implementation.
-

Functional Requirements: The Capabilities:

The Scope of the Work

The project's investigative scope covered core school operations: managing student/teacher information, attendance, grades, announcements, scheduling, and communication.

Business Data Model and Data Dictionary

- Business Data Model: Specify the core entities/classes (e.g., UML Class Model) and their relationships (e.g., a Student has 1 Parent, 1 Student has number of Grades)..
Core Classes: Student, Teacher, Admin, Parent, Grade, Attendance

Record,Announcement.

- Data Dictionary: Define the structure of the most important data flows and classes. You will define the composition of "Grade" or "Attendance Record" here.

The Scope of the Product (Use Cases):

Use Case	Actor	Short Description	Related User Story
Manage User Accounts	Admin	Add, edit, or remove user accounts.	As an admin, I want to manage user accounts.
Record Attendance	Teacher	Mark students present or absent for a class.	As a teacher, I want to record attendance.
Update Grades	Teacher	Edit and save grades for students.	As a teacher, I want to enter and edit grades.
View Grades	Student	See their grades for different subjects.	As a student, I want to view my grades.
View Attendance	Parent	Check attendance of their child.	As a parent, I want to check attendance records.
Generate Reports	Admin	Create simple attendance and grade reports.	As an admin, I want to generate attendance and grade reports.
View Announcements	Student	Read the latest school updates.	As a student, I want to see announcements.

Functional Requirements:

The system capabilities, detailed for implementation.

User Authentication and Access Control

- 1: The system shall allow users (Admin, Teacher, Student) to log in using a unique username and password.
- 2: The system shall verify user credentials before granting access.
- 3: The system shall provide distinct access rights based on the user's role.
- Fit Criterion: Attempting a Teacher function (e.g., entering grades) with a Student role login shall result in an access denial message.
- 4: The system shall allow the Admin to add, update, or remove user accounts (1 role types).

Student and Teacher Management

- 1: The system shall allow the Admin to add, view, edit, and delete student information (name, ID, class, contact details).
- 2: The system shall assign teachers to specific classes or subjects.
- 3: The system shall allow teachers to update their profile information (excluding roles or permissions).

Attendance Management

- 1: The system shall allow teachers to record attendance (Present/Absent/Excused) for each class session.
- 3: The system shall allow students to view their own attendance history.
- 4: The system shall allow parents to view their child's attendance history.

Grades Management

- 1: The system shall allow teachers to enter and update student grades for each subject.
- 2: The system shall calculate and display overall grades or averages automatically.
- 3: The system shall allow students to view their grades securely.
- 4: The system shall allow parents to view their child's grades securely.

Communication and Announcements

- 1: The system shall allow the Admin to create and send announcements to all users or selected groups (e.g., teachers, students).
- 2: The system shall display announcements on the dashboard of each relevant user.
- 3: The system shall allow teachers to post class-related notices viewable by their students.

Non-functional Requirements: The Quality of the Product :

Look and Feel Requirements

- 1 (Aesthetics): The system interface shall feature a clear, modern, and visually appealing layout.
- Fit Criterion: Post-launch usability survey shall rate the interface appeal above 4 out of 5 stars by 80% of users.

Usability and Humanity Requirements

- 1 (Efficiency of Use): The system should require the fewest possible steps to complete primary tasks (e.g., recording attendance, viewing grades).
- 2 (Guidance): The system shall provide simple labels and help text so users know exactly what to do.

- 3 (Accessibility): The system shall be accessible via standard web browsers and support different screen sizes (desktop, tablet, mobile phone browser).

Performance Requirements

- 1 (Speed/Latency): The login process and access to core dashboards shall complete quickly to prevent user frustration.
- Fit Criterion: All core page views (dashboard, grade entry, student view) shall load and render fully in less than 2 seconds for 95% of requests.
- 2 (Capacity): The system must handle simultaneous login and access by all anticipated user roles (Admin, Teachers, Students, Parents).

Security Requirements

- 1 (Integrity): The system shall store all student data securely in a central database.
- 2 (Audit): The system shall log all user activities, including login events and data updates.
- 3 (Access): The system shall restrict unauthorized access to sensitive data (e.g., a student cannot view another student's grades).

Project Issues: Conditions and Risks:

Open Issues

- Grade Weighting Logic: The final method for calculating overall grades (e.g., predefined categories, customizable weights, or total points system) needs final confirmation from school administration.

Risks

- 1 (Integration Risk): Reliance on external or historical data migration introduces a risk of data corruption or loss during transition.
- Mitigation: Conduct thorough data validation audits and implement a full rollback plan.
- 2 (Resource Risk): Insufficient user participation (especially teachers/admin) in interviews/feedback during analysis may lead to incomplete requirements.
- Mitigation: Mandate minimum participation time (e.g., attendance at prototype review sessions).

Waiting Room (Future Requirements)

The following items are out of the initial scope but should be reserved for future releases:

- Integration of School Fees or Payment Systems.
- Development of a dedicated mobile application.

Spaces

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Summary Timeline Board Calendar List Forms Goals All work Development Code Archived work items Pages Shortcuts

Search board Filter

Group

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activity diagram

Nov 30, 2025

DOZY-22

state machine diagram

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DOZY-23

crud technique

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DOZY-25

Software Requirement Specification

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